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Network Analysis

First Assignment

1 Overview

The network represents Facebook pages of politicians and the connections between them. It was taken from `https://networkrepository.com/fb-pages-politician.php`. It consists of **5,908 nodes**, each corresponding to a politician's Facebook page, and **41,729 edges**, representing the connections between these pages.
The diameter of the largest connected component is **14**, meaning the longest shortest path between any two nodes is relatively large when compared to the natural logarithm of the network size, $\ln(\mathbf{5908}) \approx \mathbf{8.7}$. This suggests that while the network is connected, some nodes are quite distant from each other.
The average clustering coefficient is **0.3551**, reflecting a strong tendency for nodes to form tightly-knit local groups. At the same time, the average shortest path length is **4.6641**, indicating that on average, any two nodes can be reached through a small number of steps. Together, these features characterize the network as exhibiting typical **small-world** properties, combining high local clustering with relatively short global distances.

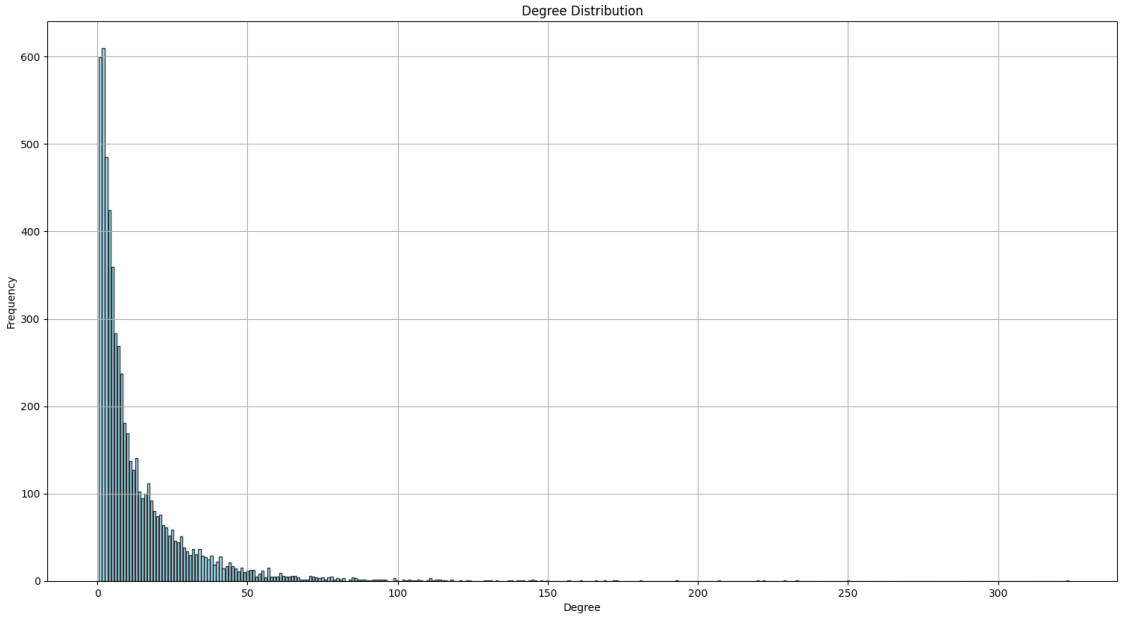


Figure 1: Degree distribution of the network.

The degree distribution shows many nodes with low degree, and the frequency of nodes decreases as the degree increases. This pattern could suggest that the network is **scale-free**. However, based on the statistical analysis from the **powerlaw** library, the network is **not clearly scale-free**. For a network to be considered scale-free, the comparison metrics must satisfy $\mathbf{R} > \mathbf{0}$ and $\mathbf{p} < \mathbf{0.05}$.
In this case, the values are $\mathbf{R} = \mathbf{16.55}$ and $\mathbf{p} = \mathbf{0.0535}$, indicating that the power-law model (our network) is **not significantly better** than the exponential alternative.
The network density is **0.002391**, meaning that only a small fraction of all possible links are present, confirming that the network is sparse, as is typical in real-world social networks.

The degree assortativity coefficient is **0.0182**, which indicates a slightly assortative mixing pattern. According to Mark Newman:

- $r > 0$: assortative
- $r < 0$: disassortative
- $r = 0$: no degree correlation

In our case, a small positive r suggests a weak tendency of nodes to connect with others of similar degree, which aligns with the typical structure of social networks.

Top 10 Nodes by Degree and Betweenness Centrality

Rank	Degree	Betweenness Centrality
1	Manfred Weber (323)	Barack Obama (0.2683)
2	Joachim Herrmann (250)	Manfred Weber (0.0555)
3	Barack Obama (233)	Angela Merkel (0.0541)
4	Katarina Barley (229)	Hillary Clinton (0.0483)
5	Arno Klare MdB (222)	Justin Trudeau (0.0451)
6	Katja Mast (220)	Sir Peter Bottomley MP (0.0373)
7	Angela Merkel (207)	Malcolm Turnbull (0.0354)
8	Martin Schulz (193)	Narendra Modi (0.0341)
9	Niels Annen (181)	Loïc Hervé (0.0318)
10	Heike Baehrens (173)	Boris Johnson (0.0290)

Table 1: Top 10 nodes by Degree and Betweenness Centrality.

The nodes listed in the **Degree** column are highly connected and likely serve as central actors in the information flow within the network. On the other hand, the nodes listed in the **Betweenness Centrality** column play a crucial role in connecting different communities. They often act as bridges between otherwise disconnected parts of the network. The removal of such nodes could lead to structural holes, reducing or even breaking communication between communities.

2 Community Analysis

For this part, I used three different community detection algorithms, selected based on the characteristics of my network:

- **Louvain**
- **Greedy Modularity**
- **Label Propagation**

Here are the results obtained from each algorithm:

Algorithm	Communities Detected	Execution Time (s)
Label Propagation	240	0.2147
Louvain	30	0.5473
Greedy Modularity	36	17.5361

Table 2: Number of communities detected and execution time for each algorithm.

As expected, the **Label Propagation** algorithm, while very fast, tends to produce a large number of small communities, often composed of just 2 or 3 nodes, due to its purely local update mechanism.
In contrast, **Louvain** and **Greedy Modularity**, both based on modularity optimization, resulted in fewer but more balanced communities, which are generally more suitable for interpretation in larger networks.

3 Community Quality Analysis

To evaluate the similarity between the community structures obtained by different algorithms, I computed the **Normalized Mutual Information (NMI)** and the **Adjusted Rand Index (ARI)** for each pair of partitions:

First Algorithm	Second Algorithm	NMI	ARI
Louvain	Greedy Modularity	0.7548	0.4663
Louvain	Label Propagation	0.7956	0.5392
Greedy Modularity	Label Propagation	0.6811	0.3154

Table 3: Comparison of community detection algorithms using NMI and ARI.

The NMI values, **all above 0.68**, indicate that the three algorithms **share a substantial amount of information** despite differences in the number of detected communities. In particular, the **Label Propagation** method, which yields a much higher number of small communities, still aligns well with the others in terms of general structure. This suggests that all methods **capture a similar underlying organization** of the network. On the other hand, the ARI values are lower, which is expected: ARI is more sensitive to exact cluster assignments, and the differing number and size of communities across methods naturally lead to more variation. Nevertheless, the **positive ARI values** show that the methods **still produce comparable results**, particularly between **Louvain and Label Propagation**.

4 Communities Overview

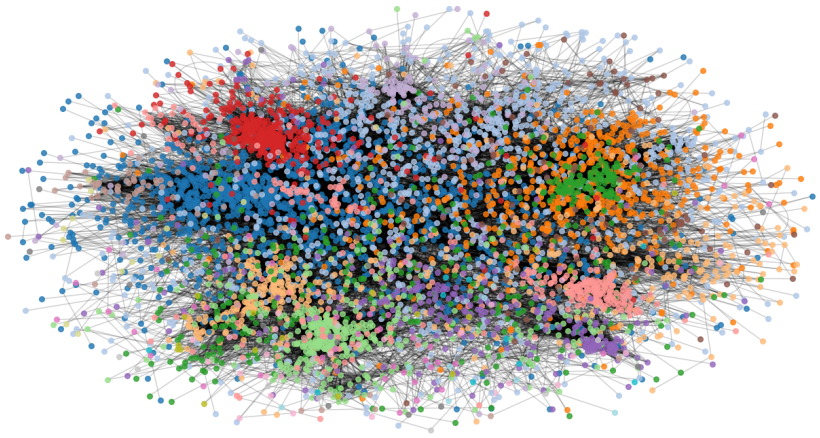


Figure 2: Greedy Modularity



Figure 3: Label Propagation

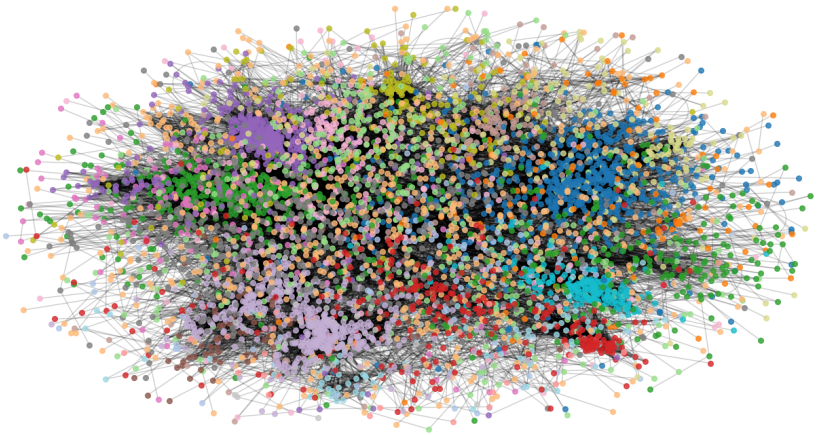


Figure 4: Louvain